8.1 The Dynamic Rewards System of Bitcoin Mining: Block Rewards, Transaction Fees, and Halvings

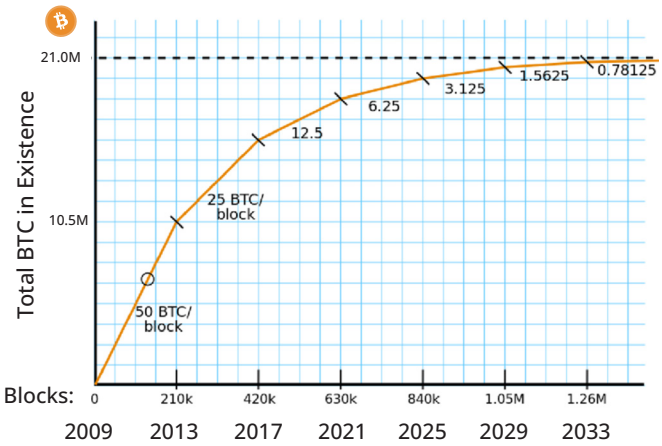
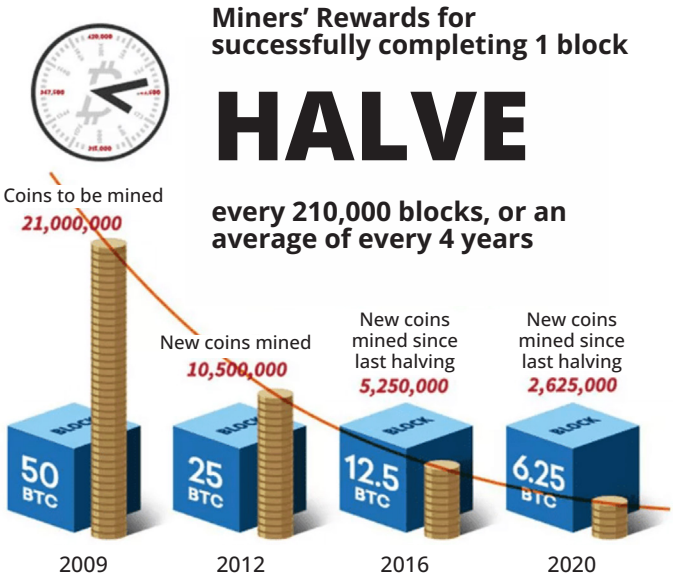
Satoshi Nakamoto, the creator of ***Bitcoin***, came up with a smart solution for distributing new ***bitcoin*** via a block reward system in a decentralized manner.

The **bitcoin supply schedule** is his plan for the creation and release of new bitcoin into circulation, which is designed to maintain the scarcity of bitcoin over time.

In the early days of ***Bitcoin***, miners were awarded 50 ***bitcoin*** for each block they mined. This block reward serves as a financial motivation for miners to invest in powerful hardware and electricity for their mining operations.

However, to control the supply of new ***bitcoin*** and maintain stability in the network, the block reward **halves** approximately every 210,000 blocks. This process, known as "**halving**," reduces the amount of new ***bitcoin*** released into circulation and continues to incentivize miners to secure the network and ensures its decentralization.

- Let's say you have a jar that can only hold 1000 pieces of candy. Every day, you get to add 10 pieces of candy to the jar. This is how new ***bitcoin*** are created and added to the supply through the process of mining. However, after every four years, the amount of candy you can add to the jar is cut in half.

This is like the **halving** of ***Bitcoin***, which **slows down the rate** at which new bitcoin are created. The ultimate goal is to maintain scarcity and limit **the total number of *bitcoin* to 21 million units**, just like the jar can only hold a limited amount of candy.

Bitcoin Supply Schedule